Supratik Sarkar | Curriculum Vitae

Joint Quantum Institute, University of Maryland
4196 Stadium Dr, College Park, MD 20742

sarkar@umd.edu +1 (240)610-7226 Google Scholar Website

(Last updated March 2026)

Education

Joint Quantum Institute, University of Maryland **College Park, MD, USA**
PhD in Electrical and Computer Engineering *2021 - 2026*
GPA: 3.9/4
Supervisor: Prof. Mohammad Hafezi

Institute for Quantum Computing, University of Waterloo **Waterloo, ON, Canada**
MASc in Electrical and Computer Engineering (Quantum Information) *2019 - 2021*
GPA: 96.20/100
Thesis: "Creating and shaping light at single photon level"
Supervisor: Prof. Michal Bajcsy

Jadavpur University **Kolkata, WB, India**
Bachelor of Engineering (B.E.), Electronics and Tele-communication Engineering *2014 - 2018*
GPA: 9.26/10 (First class with honors)

Research interests

- Photonic integrated circuits
- Atomic, molecular, and optical physics
- Correlated matter and many-body physics
- Quantum information and quantum computing

Technical skills

- **Optics**
  Free-space and fiber optical systems, real-/Fourier-space imaging, ultra-fast spectroscopy, correlation measurements, interferometry, optics in cryogenic environments, automating opto-electronic experiments
- **Basic nanofabrication**
  Electron beam lithography, physical vapor deposition, reactive ion etching, spin coating, reflectometry, wet bench
- **Programming languages and softwares**
  C, C++, C#, Python, SQL, Verilog, MATLAB, Mathematica, ANSYS Lumerical, Zemax, Tidy3D, COMSOL, CAD tools, KLayout, GDSFactory, LabVIEW, Qiskit, Qutip, Tensorflow, LaTeX

Research experience

PhD research

- **Topological photonics for engineering nonlinear interactions and photonic computing**
  - Designed photonic integrated circuits (silicon nitride, lithium niobate) using both *ab initio* analytical techniques, and commercial software like ANSYS Lumerical, Zemax, COMSOL, and Tidy3D.
  - Performed tape-out and characterization of said devices for engineering optical nonlinearity for frequency comb generation and photonic computing.

- **Optically probing strongly correlated electronic states in 2D semiconductors**
  - Experimentally using tools from quantum optics and light-matter interaction to explore many-body physics, topology, and chirality in photonic and condensed matter systems, particularly two-dimensional semiconductors like transitional metal dichalcogenides, aimed towards understanding exotic phases of matter.
  - Designed, built, maintained optical setups (free-space and fiber optics) and electronics and automated experiments for real-/momentum-space imaging, ultra-fast spectroscopy, non-linear optics, interferometry, and time-tagged measurements in two 4K cryostats (Attodry, Montana), and one dilution fridge (Bluefors).
  - Designed emitter-coupled hybrid photonic devices using both *ab initio* analytical techniques, and commercial software like ANSYS Lumerical.
  - Collaborated with multiple research groups and presented my research at multiple international conferences.
  - Trained multiple new students on the nanophotonic simulations and our cryogenic experimental setups.

Masters research

- **Deterministic single photon subtraction for engineering exotic non-classical states of light**
  - Performed analytical and numerical analysis to propose methods of deterministic single photon subtraction from any arbitrary state of light using a single quantum emitter coupled with a chiral waveguide or a cavity.
  - Demonstrated how this can be used to engineer non-classical exotic states of light that have applications in quantum information theory and metrology.

- **Designing optical antennas with Inverse Design for enhanced directional emission from NV centers**
  - Used adjoint optimization and inverse design techniques to engineer diamond nanophotonic devices that allow for highly directional emission of radiation from nitrogen-vacancy (NV) centers compared to traditional techniques.
  - Designed and built an experimental setup for scanning confocal microscopy of the proposed device.

Undergraduate research

- **Bachelor's Degree Project: Tunneling in Graphene SymFETs** (*2017 - 2018*)
  - **Supervisor**: Prof. Chayanika Bose, Jadavpur University

- **Single Photon Subtraction from Continuous Variable Cluster States** (*2017*)
  - **Supervisors**: Prof. Nicolas Treps, and Prof. Valentina Parigi, Multimode Quantum Optics Group, Laboratoire Kastler Brossel (CNRS, ENS, UPMC-Sorbonne)

- **Influence of Polarization and Self-polarization Charges on Impurity Binding Energy in Spherical Quantum Dot with Parabolic Confinement** (*2016 - 2017*)
  - **Supervisor**: Prof. Chayanika Bose, Jadavpur University

- **Detection of Celiac Disease from Endoscopy Images Using Convolutional Neural Networks** (*2016*)
  - **Supervisor**: Prof. Debdoot Sheet, Indian Institute of Technology Kharagpur

- **True Random Number Generators** (*2015*)
  - **Supervisor**: Prof. Subhamoy Maitra, Indian Statistical Institute, Kolkata

Teaching assistantship

- **ECE 404: Geometric and Physical Optics** *Winter semester, 2020*

- **Instructor**: Prof. Michal Bajcsy, University of Waterloo
- Held weekly office hours, and graded assignments for a class of roughly 50 undergraduate students.

Manuscripts

* Authors contributed equally

Under peer-review

1. L. Xu*, A. Padhye*, **Supratik Sarkar**, A. Parhizkar, C. J. Flower, G. Moille, K. Srinivasan, M. Hafezi, M. J. Mehrabad. **Single-shot realization of 10000-mode octave-spanning artificial gauge fields.** (2026).
2. A. Pasharavesh*, **Supratik Sarkar***, G. Bappi, J. Du, S. S. Venuturumilli, M. Bajcsy. **Photon subtraction using a three-level emitter coupled to a chiral waveguide.** (2026).
3. B. Gao, M. Ghafariasl, M. J. Mehrabad, T.-S. Huang, L. Zhang, D. Session, P. Upadhyay, R. Ma, G. Alshalan, D. G. Suárez-Forero, **Supratik Sarkar**, S. Park, H. Jang, K. Watanabe, T. Taniguchi, M. Xie, Y. Zhou, M. Hafezi. **Probing quantum anomalous Hall states in twisted bilayer WSe_2 via attractive polaron spectroscopy**. arXiv:2504.11530 (2025).

Peer-reviewed publications

1. B. Semnani, S. S. Venuturumilli, M. Soltani, A. Pasharavesh, N. Videnov, P. Adhikary, P. Anderson, **Supratik Sarkar**, V. R. R. Muthu, M. Bajcsy. **Probing individual quantum emitters in bulk semiconductors via photonic nanojets.** Science Advances (2026).
2. P. Upadhyay*, D. G. Suárez-Forero*, T.-S. Huang*, M. J. Mehrabad, B. Gao, **Supratik Sarkar**, D. Session, K. Watanabe, T. Taniguchi, Y. Zhou, M. Knap, M. Hafezi. **Giant enhancement of exciton diffusion near an electronic Mott insulator**. Science 391, 394-398 (2026).
3. M. J. Mehrabad*, L. Xu*, G. Moille, C. J. Flower, **Supratik Sarkar**, A. Padhye, S.-C. Ou, D. G. Suárez-Forero, M. Ghafariasl, Y. Chembo, K. Srinivasan, M. Hafezi. **Multi-timescale frequency–phase matching for high-yield nonlinear photonics**. Science 390, 612-616 (2025).
4. **Supratik Sarkar***, M. J. Mehrabad*, D. G. Suárez-Forero*, L. Gu*, C. J. Flower, L. Xu, K. Watanabe, T. Taniguchi, S. Park, H. Jang, Y. Zhou, M. Hafezi. **Sub-wavelength optical lattice in 2D materials**. Science Advances 11, eadv2023 (2025). [*Featured on Science Advances website*]
5. D. G. Suárez-Forero*, R. Ni*, **Supratik Sarkar***, M. J. Mehrabad*, E. Mechtel, V. Simonyan, A. Grankin, K. Watanabe, T. Taniguchi, S. Park, H. Jang, M. Hafezi, You Zhou. **Chiral flat-band optical cavity with atomically thin mirrors**. Science Advances 10, eadr5904 (2024). [*Featured on Science Advances website*]
6. B. Gao*, D. G. Suárez-Forero*, **Supratik Sarkar***, T.-S. Huang, D. Session, M. J. Mehrabad, R. Ni, M. Xie, P. Upadhyay, J. Vannucci, S. Mittal, K. Watanabe, T. Taniguchi, A. Imamoglu, Y. Zhou, M. Hafezi. **Excitonic Mott insulator in a Bose-Fermi-Hubbard system of moiré WS_2/WSe_2 heterobilayer**. Nature Communications, 15, 2305 (2024).
7. M. Walschaers, **Supratik Sarkar**, V. Parigi, and N. Treps. **Tailoring Non-Gaussian Continuous-Variable Graph States**. Physical Review Letters 121, 220501 (2018).
8. **Supratik Sarkar**, S. Sarkar, and C. Bose. **Influence of polarization and self-polarization charges on impurity binding energy in spherical quantum dot with parabolic confinement**. Physica B: Condensed Matter, 541, 75-78 (2018).

Thesis

1. **Supratik Sarkar**. **Creating and shaping light at single photon level**. Master's thesis, University of Waterloo, http://hdl.handle.net/10012/17024 (2021).
2. **Supratik Sarkar**, and S. Sarkar. **Tunneling in Graphene SymFETs**. Bachelor's project report, Jadavpur University, arXiv: 1805.09659 (2018).

Patent applications

1. M. J. Mehrabad, L. Xu, A. Padhye, **Supratik Sarkar**, M. Hafezi. **Systems of ultrabroadband multimodal artificial gauge fields.** Provisional US patent (Filed 2026)
2. M. J. Mehrabad, L. Xu, **Supratik Sarkar**, Z.-Y. Wei, M. Hafezi. **Topological photonics architectures for optical computing and artificial intelligence.** Provisional US patent (Filed 2026)
3. **Supratik Sarkar**, M. J. Mehrabad, Y. Zhou, M. Hafezi. **Systems and methods for an active nanoplasmonic platform for controllable light-matter interaction.** Provisional US patent (Filed 2025)

Media features

1. **Researchers Identify Groovy Way to Beat Diffraction Limit** - Joint Quantum Institute news
2. **With Passive Approach, New Chips Reliably Unlock Color Conversion** - Joint Quantum Institute news, Phys.org, Optics.org, MIT Technology Review China
3. **Sudden Breakups of Monogamous Quantum Couples Surprise Researchers** - Joint Quantum Institute news, Phys.org

Conferences and talks

1. **Symposium II: New trends in nanophotonics and advanced materials**. 15th International Conference on Metamaterials, Photonic Crystals and Plasmonics (META). Malaga, Spain (2025). [*Invited talk*]
2. **Sub-wavelength optical lattice in 2D materials**. Conference on Lasers and Electro-Optics (CLEO). Long Beach, CA, USA (2025). [*Contributed talk*]
3. **Sub-wavelength optical lattice in 2D materials**. International Conference on Physics of Excitons and Polaritons in Semiconductors (PEPS). Reykjavík, Iceland (2024). [*Contributed talk + Poster*]
4. **Chiral Optical Nano-Cavity with Atomically Thin Mirrors**. NANOMETA. Seefeld in Tirol, Austria (2024). [*Breakthrough talk*]
5. **Shot noise measurements for light-induced superconductivity**. Dr. Milan Allan's group. Leiden University, Netherlands (2023).
6. **Creating non-classical states using deterministic single photon subtraction in waveguides and bimodal cavities**. APS DAMOP (2021). [*Contributed talk*]
7. **Deterministic single photon subtraction for engineering exotic non-classical states of light**. APS March Meeting (2020). [*Poster presentation*]
8. **Deterministic single photon subtraction using solid state emitters coupled to chiral waveguides**. Frontiers in Optics + Laser Science APS/DLS. Washington DC, USA (2019). [*Poster presentation*]
9. **Deterministic Single Photon Subtraction**. Waterloo-HKUST Workshop on Emerging Quantum Technologies in Solid-State and Atomic System Workshop (*2019*). [*Poster presentation*]
10. **Deterministic Single Photon Subtraction, and Antenna Model for Metasurface-Assisted Enhancement of Light-Matter Interaction**. Celebratory Waterloo-Rochester Photonics Symposium (*2019*). [*Poster presentation*]

Awards, scholarships & grants

1. Invention of the year award finalist in quantum category, given by University of Maryland (*2026*)
2. Outstanding Graduate Assistant Award, given by University of Maryland (*2026*)
3. Nanophotonics (Wiley) poster award, given by European Physical Society (EPS) Nanometa (*2026*)
4. NSF I-Corps, given by the National Science Foundation (NSF) (2025)
5. Dean's Fellowship, given by University of Maryland (*2021*)
6. WIN Nanofellowship, given by Waterloo Institute for Nanotechnology (*2020*)
7. Faculty of Engineering (FOE) Award, given by the Department of ECE, University of Waterloo for academic and research performance (*2020*)
8. International Masters Student Award, given by the University of Waterloo (*2019*)
9. Graduate Research Studentship (GRS) funding, given by the University of Waterloo (*2019*)

10. Charpak Masters Scholarship, given by the Govt. of France to pursue Masters in France (Declined) (*2018*)
11. CNRS internship scholarship for an internship at Laboratoire Kastler Brossel (LKB), France (*2017*)
12. Certificate of Merit, given by Minister of Human Resource Development, Govt. of India for being among the top 0.1% of candidates in Biology in All India Senior School Certificate Examination (*2014*)

Journal referee activities

- American Physical Society journals - Physical Review Letters, Physical Review Applied
- American Chemical Society journals - Nano Letters

Available references

- Prof. Mohammad Hafezi
- Prof. You Zhou
- Prof. Daniel G. Suárez-Forero
- Prof. Michal Bajcsy